

QuantMesh x402

Atomic Multi-Agent Service Router

Problem Statement: PS0404

Algorand AVM Micropayments

- **Team:** QuantMesh Innovators
- **College:** CSBS Dept. BV(DIU)COE



1. Problem Statement & Motivation



Subscription Fatigue

- Crypto signal APIs cost **\$100–\$500/month**.



Single-AI Risk

- Single AI models hallucinate or misread



Payment Friction

- Executing multi-agent workflows requires

2. Proposed Solution / Idea

⚡ QuantMesh x402 Architecture

- **Pay-Per-Signal: \$0.007 USDC** per execution via Algorand x402 AVM protocol.
- **🛡️ Pre-Execution Zero-Fee Guarantee:** Hono router validates all 4

🔑 Killer Innovation: Zero-Fee Shield

Client Request -> Hono Gateway

- Run Worker A (FinBERT NLP)
- Run Worker B (Whale Flow)

3. Identified Paying User & Business Model



Autonomous AI Agents

- Agentic trading scripts & DeFi bots.



Quant Traders & Analysts

- Retail traders wanting high-confidence



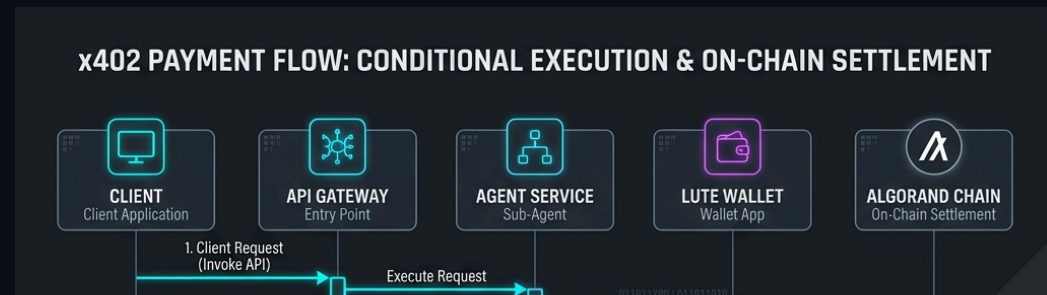
Business Model

- **\$0.007 USDC / Execution**
- **\$0.005** to sub-agent liquidity

4. x402 Payment Flow (Technical Core)

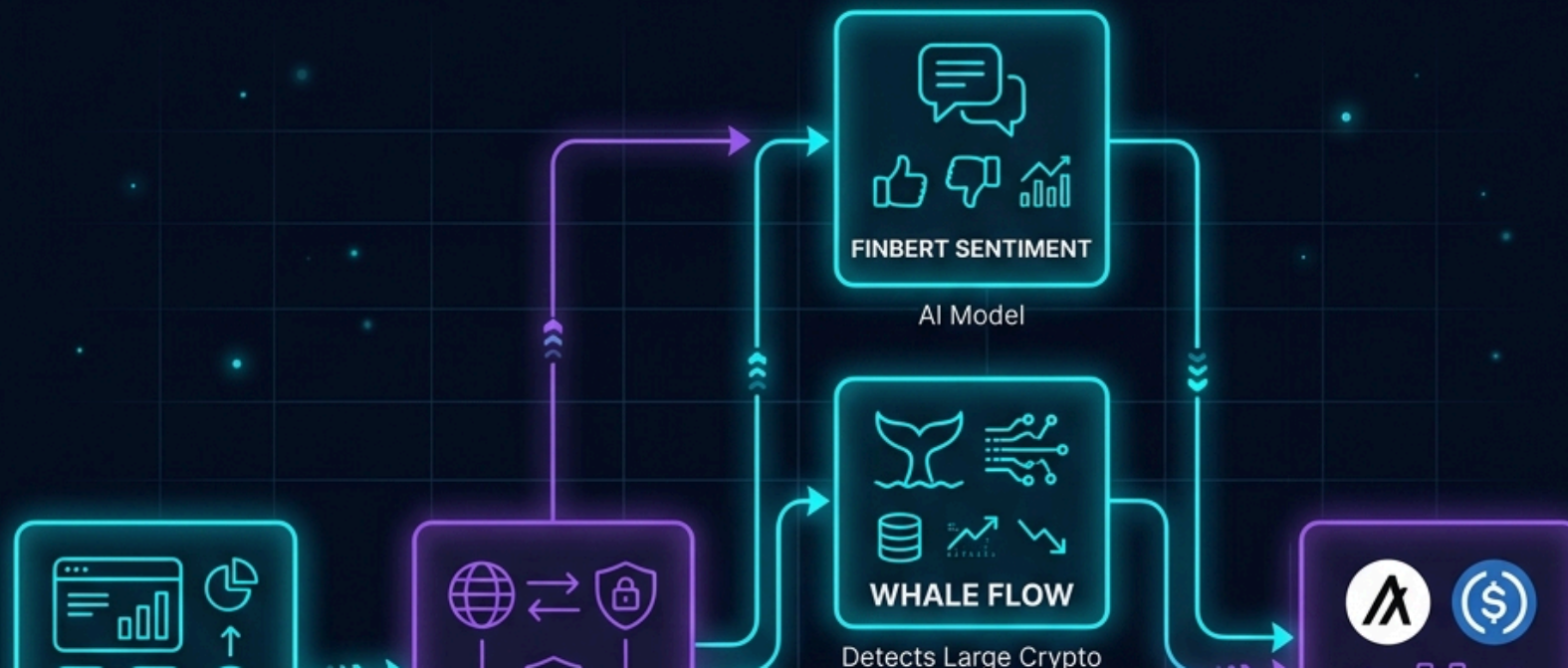
The 4-Step AVM Protocol

1. **Challenge (402):** Server validates sub-agents, then returns **HTTP 402 Payment Required** + payment request



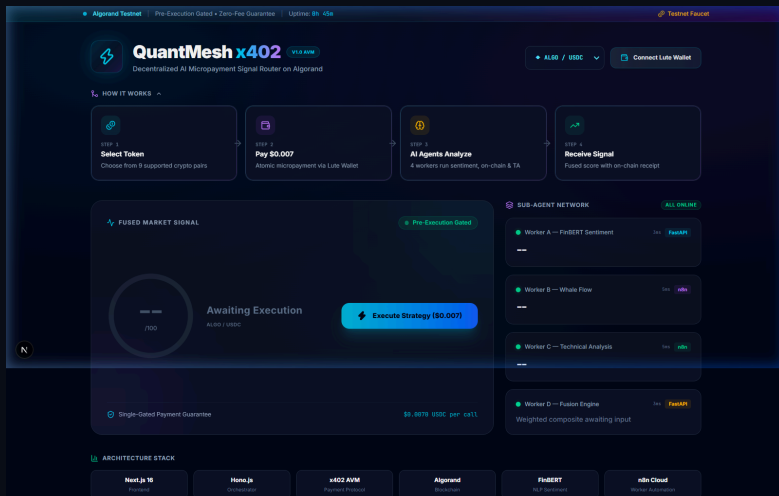
5. System Architecture & Tech Stack

WEB3 AI DATA FLOW ARCHITECTURE

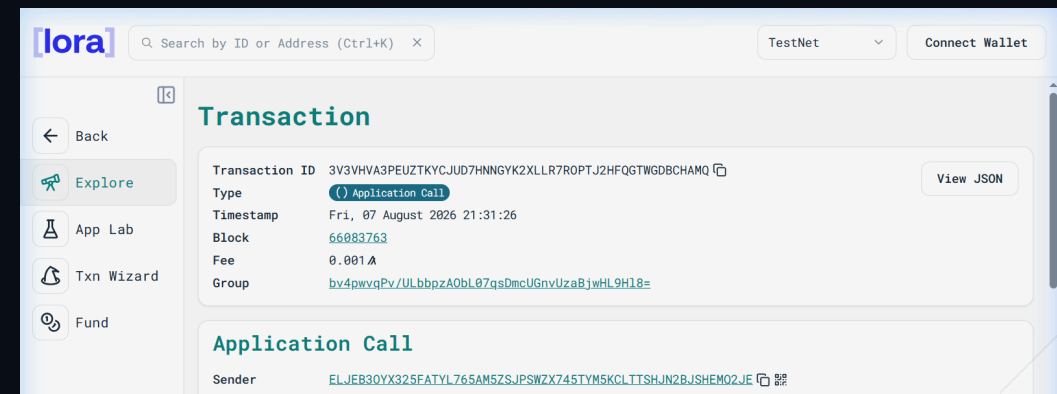


6. Live Demo & On-Chain Proof (Lora Explorer)

Next.js 16 Live Dashboard



Lora Testnet Tx & Group Proof

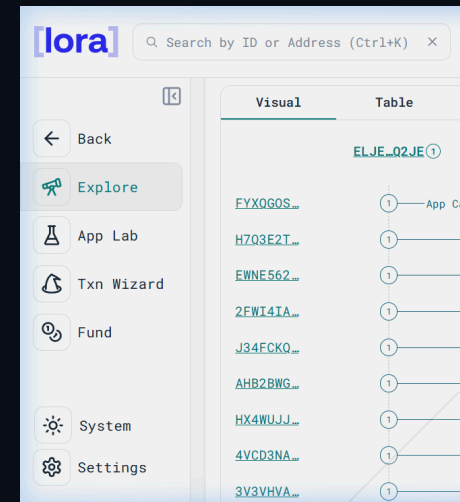


6b. Atomic Group Transaction

Visual Flow

⚡ Algorand AVM Group Flow

- **Atomic Settlement:** All sub-agent verifications and micropayments execute atomically within an AVM transaction group.
- **Group Hash:**
`bv4pwvqPv/ULbbpzA0bL07qsDmcUGnvUzaBjwHL9H18=`
- **Zero Wasted Gas:** If any single tx in the group



7. Innovation & Competitive Differentiation

Feature / Criteria	Traditional DeFi APIs	Generic x402 Routers	QuantMesh x402 (PS0404)
Payment Model	\$100–\$500/mo SaaS	Pay per request upfront	Pay-Per-Execution (\$0.007)
Failure Protection	✗ Money lost if API fails	✗ Charged before execution	🛡️ Pre-Execution Zero-Fee Shield

8. Completeness & Functionality

✓ **Live & Operational (Done)**

- **Hono Orchestrator**
deployed live on

9. Real-World Impact & Scalability



Democratic Access

- Enables micro-investors and



Mainnet Readiness

- **100% Ready:** Requires changing only 2 environment variables:
 - `ALGORAND_NETWORK=mainnet`
 - `USDC_ASA_ID=315667040`



Scalability

- Hono + AWS EC2 handles **1,000+**



Summary: Why QuantMesh x402 Wins

- **Solves PS0404:** True Agentic Payments with Algorand AVM x402 protocol.
- **Consumer Protection:** Pre-execution gating guarantees \$0 wasted fees.
- **Institutional Signal:** Fuses sentiment NLP + on-chain whale flow + technical indicators.
- **Production Live Today:** Live EC2 API + Next.js Dashboard + Verified On-Chain Proof.